Topic 8 — Maximum Likelihood Estimation

The Likelihood Function

Suppose we believe data $x_1, \dots, x_n$ is the output of i.i.d. random variables $X_1, \dots, X_n \sim X$, where X depends on an unknown parameter $\theta \in \mathbb{R}$.

Definition

The **likelihood** (or **likelihood function**) of the data, given θ, is

$$\mathcal{L}(\theta) = \mathcal{L}(\theta \mid x_1, \dots, x_n) := \mathbb{P}(X_1 = x_1, \dots, X_n = x_n \mid \theta).$$

Since $X_1, \dots, X_n$ are independent,

$$\mathcal{L}(\theta \mid x_1, \dots, x_n) = \mathbb{P}(X_1 = x_1 \mid \theta) \cdots \mathbb{P}(X_n = x_n \mid \theta).$$

Example: Tossing a Coin

Example

A coin has $\mathbb{P}(H) = \theta \in [0,1]$, unknown. Tossed $n = 10$ times, data $= (H,H,T,H,T,H,H,T,H,T)$:

$$\mathbb{P}(\text{data} \mid \theta) = \theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot (1-\theta) \cdots = \theta^6(1-\theta)^4 =: \mathcal{L}(\theta \mid \text{data}).$$

In general, if the data contains k heads (and $n-k$ tails),

$$\mathcal{L}(\theta \mid \text{data}) = \theta^k(1-\theta)^{n-k}.$$

Maximum Likelihood Estimation

Definition

The **maximum likelihood estimate** θ_* of θ is the argument of the maximum of the likelihood function:

$$\theta_* = \operatorname*{argmax}_\theta \mathcal{L}(\theta), \qquad \text{i.e.} \qquad \mathcal{L}(\theta_*) = \max_\theta \mathcal{L}(\theta).$$

Remark

The standard approach is the **log-likelihood function** $L(\theta) := \ln \mathcal{L}(\theta)$. Since $\ln x$ is increasing, $\mathcal{L}(\theta) \leq \mathcal{L}(\theta_*) \iff L(\theta) \leq L(\theta_*)$, so θ_* maximises both $\mathcal{L}$ and L.

Remark

To check θ_* is a maximum of $L(\theta)$: it is enough that $L'(\theta_*) = 0$ and $L''(\theta_*) < 0$.

General algorithm: write $L(\theta) = \ln \mathbb{P}(X_1 = x_1 \mid \theta) + \cdots + \ln \mathbb{P}(X_n = x_n \mid \theta)$; solve $L'(\theta) = 0$ for θ_*; check $L''(\theta_*) < 0$.

Example: MLE for the Coin (Bernoulli)

Example

$$L(\theta) = \ln \mathcal{L}(\theta) = k \ln \theta + (n-k) \ln(1-\theta).$$

$$L'(\theta) = \frac{k}{\theta} - \frac{n-k}{1-\theta} = \frac{k - n\theta}{\theta(1-\theta)}.$$

$L'(\theta) = 0 \iff \theta = \dfrac{k}{n}.$

Example

$$L''(\theta) = -\frac{k}{\theta^2} - \frac{n-k}{(1-\theta)^2} < 0 \quad \text{for all } \theta \in [0,1].$$

Hence $\theta_* = \dfrac{k}{n}$ is the maximum likelihood estimate.

Connection to the Law of Large Numbers

Remark

Let $H \leftrightarrow 1$, $T \leftrightarrow 0$: $X_i \sim \text{Bernoulli}(\theta)$, $\mathbb{E}(X_i) = \theta$. Let $S_n = X_1 + \cdots + X_n \sim Bin(n, \theta)$ be the number of heads; then $\bar{X}_n = \dfrac{S_n}{n} = \dfrac{k}{n} = \theta_*$. By LLN, $\bar{X}_n \to \mathbb{E}(X) = \theta$ as $n \to \infty$.

So the maximum likelihood estimate is exactly the sample average here — and it converges to the true θ as the sample size grows. More generally, it can be shown that if $\theta_{*,n}$ is the MLE from a sample of size n, then $\lim_{n\to\infty} \theta_{*,n} = \theta_*$, the true parameter.

Example: A Discrete Distribution with Unknown Parameter

Example

Suppose $X : \Omega \to \{0, 1, 2, 3, 4\}$ has PMF

X	0	1	2	3	4
$\mathbb{P}$	$\frac{3\theta}{5}$	$\frac{\theta}{5}$	$\frac{\theta}{5}$	$\frac{2(1-\theta)}{5}$	$\frac{3(1-\theta)}{5}$

(note the probabilities sum to 1 for any $\theta \in [0, 1]$). Given the sample $0, 1, 0, 3, 4, 0, 1, 1, 2, 1, 4, 4, 2$ (three 0's, four 1's, two 2's, one 3, three 4's), find θ_*.

Example

$$\mathcal{L}(\theta) = (\mathbb{P}(X = 0))^3 (\mathbb{P}(X = 1))^4 (\mathbb{P}(X = 2))^2 \mathbb{P}(X = 3)(\mathbb{P}(X = 4))^3 = C\theta^9(1 - \theta)^4,$$

where $C = \left(\frac{3}{5}\right)^3 \left(\frac{1}{5}\right)^4 \left(\frac{1}{5}\right)^2 \frac{2}{5} \left(\frac{3}{5}\right)^3 > 0$ is a constant.

Example

$$L(\theta) = \ln C + 9 \ln \theta + 4 \ln(1 - \theta), \qquad L'(\theta) = \frac{9}{\theta} - \frac{4}{1 - \theta}.$$

$$L'(\theta) = 0 \implies 9(1 - \theta) = 4\theta \implies \theta = \frac{9}{13} \in [0, 1].$$

$$L''(\theta) = -\frac{9}{\theta^2} - \frac{4}{(1 - \theta)^2} < 0 \quad \text{for all } \theta.$$

Hence $\theta_* = \frac{9}{13}$.

Example: Poisson MLE

Example

Let $k_1, \dots, k_n$ be a sample of i.i.d. $X_i \sim Po(\lambda)$, $\lambda > 0$ unknown, $\mathbb{P}(X = k) = \dfrac{\lambda^k}{k!} e^{-\lambda}$.

$$\mathcal{L}(\lambda) = \frac{1}{k_1! \cdots k_n!} \lambda^{k_1 + \cdots + k_n} e^{-\lambda n} = C \lambda^{k_1 + \cdots + k_n} e^{-\lambda n}.$$

Example

$$L(\lambda) = \ln C + (k_1 + \cdots + k_n) \ln \lambda - \lambda n, \qquad L'(\lambda) = \frac{k_1 + \cdots + k_n}{\lambda} - n.$$

$$L'(\lambda) = 0 \implies \lambda_* = \frac{k_1 + \cdots + k_n}{n} \qquad \text{(the sample mean)}.$$

$$L''(\lambda) = -\frac{k_1 + \cdots + k_n}{\lambda^2} < 0 \quad \text{for all } \lambda.$$

So λ_* is the maximum likelihood estimate — the sample mean, as expected.

Remark: Continuous Random Variables

Remark

For continuous random variables we cannot use $\mathbb{P}(X_i = x_i)$, since each such probability is 0. Instead, we use the **PDF**:

$$\mathcal{L}(\theta \mid x_1, \dots, x_n) = f_X(x_1 \mid \theta) \cdots f_X(x_n \mid \theta),$$

where $f_X(x \mid \theta)$ is the PDF of X at x (which depends on θ). Everything else — log-likelihood, $L' = 0$, $L'' < 0$ — carries over unchanged.